

Syllabus

Spring semester 2026

Course Title	General Biology 2		Course Code	CB3203		Credit	4	Classroom	Chang 408
			Type			Hours	4		
Teacher	Name	Class no.	Office	Office hours		Contact		E-mail	
	Ahn, Jung Hoon	1	Chang 403	Mon 5 Wed 5 Fri 5		606-2335		hoony@kaist.ac.kr	

1. Course Description

Aims and objectives	■ Understand and explain the principles and concepts of college-level biology <i>in English</i>
	■ Understand evolution at the molecular level and phylogenetic reconstruction
	■ Compare the key characteristics of viruses, prokaryotes, plants and animals
	■ Integrate and explain the various parts of animal anatomy and physiology,

2. Core Target Competencies

Subject Core Competencies				
1. Analytical Thinking	2. Creative Thinking	3. Leadership and Teamwork	4. Digital Literacy	5. Habits/ Mindset
Students will learn biological theories and principles mainly through self-directed learning and group discussion in class. The biological contents that are covered in this subject can be divided into three main areas: (1) molecular evolution, (2) viruses and representative living organisms, and mainly (3) animal anatomy and physiology. (1) The diversity of life - Phylogenetic reconstruction - Viruses - Prokaryotes (2) Animal anatomy and physiology - Digestive systems - Transport and Excretory systems - Development - Defenses and Neuronal signals				

3. Textbooks and References

Text : Campbell et al., Biology, A Global Approach, 11th/12th edition, Pearson

Workbook : A problem-solving workbook is provided.

4. Class Method

Lecture / Individual or Group Worksheet / Group presentation and discussion

5. Evaluation

A. Evaluation Criteria

Team Presentation ¹	Workbook ²	Midterm exam ³	Final exam ³	Attendance and attitude ⁴	Total
10%	20%	30%	30%	10%	100%

B. Evaluation Method

1. Team presentation about the topic of your choice in each chapter. Minimize book content and present a content out of the book. Choose some topic that you have any interest in, deepen your understand in the topic and explain your understand to persuade students in the class.
2. Workbook will be given for your self-study. Use it for your note taking.
3. Exams will contain short answer questions and essay questions.
4. Absence without notice: -1pt, being late (every twice : -0.5pt)

6. Criteria for Awarding Credits

☞ Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

7. Weekly Plans

Week (period)	Topic and Contents	Class activity
1 (2/23~)	Introduction to the course Ch. 22 Phylogenetic Reconstruction	
Core Competencies	1. Analytical Thinking / 2. Creative Thinking	
2 (3/2~)	Ch. 22 Phylogenetic Reconstruction	What is the purpose of creating a phylogenetic tree?
Core Competencies	1. Analytical Thinking / 2. Creative Thinking	
3 (3/9~)	Ch. 40 The animal body	Why big animal has high BMR?
Core Competencies	3. Leadership and Teamwork	
4 (3/16~)	Ch. 41 Chemical Signals in Animals	Does hormone change mood?
Core Competencies	3. Leadership and Teamwork	
5 (3/23~)	Ch. 42 Animal Digestive Systems	What should we eat for health?
Core Competencies	4. Digital Literacy	
6 (3/30~)	Ch. 43 Animal Transport Systems	Is slow breathing good for health?
Core Competencies	5. Habits/ Mindset	
7 (4/6~)	Ch. 44 Animal Excretory Systems	Why is the stone formed in bladder and how is it cured?
Core Competencies	5. Habits/ Mindset	
8 (4/13~)	Midterm exam (4/14~4/17)	

Week (period)	Topic and Contents	Class activity
9 (4/20~)	Ch. 27 Prokaryotes	How can bacteria live in extreme condition?
Core Competencies	1. Analytical Thinking / 2. Creative Thinking	
10 (4/27~)	Ch. 45 Animal Reproductive system	How to give birth to a child of the desired sex
Core Competencies	1. Analytical Thinking / 2. Creative Thinking	
11 (5/4~)	Ch. 46 Development in Animals	How can we make stem cells by expressing a few genes in the cells?
Core Competencies	3. Leadership and Teamwork / 5. Habits/ Mindset	
12 (5/11~)	Ch. 47 Animal Defenses Against Infection	How can we develop guided missile to attack cancer cells?
Core Competencies	3. Leadership and Teamwork / 5. Habits/ Mindset	
13 (5/18~)	Ch. 48 Electrical Signals in Animals	How is Lou Gehrig diseases caused and cured?
Core Competencies	4. Digital Literacy	
14 (5/25~)	Ch. 49 Neural Regulation in Animals	How does dopamine regulate our desire?
Core Competencies	5. Habits/ Mindset	
15 (6/1~)	Ch. 26 Introduction to Viruses	How can we cope with Corona virus?
Core Competencies	5. Habits/ Mindset	
16 (6/8~)	Complementary classes	
17 (6/15~)	Final Exams (6/15~6/19)	

※ The above plans may change according to the academic schedule.